Invertebrates are animals that lack a backbone. They occupy almost every habitat on Earth, from the scalding water released by deep-sea hydrothermal vents to the frozen ground of Antarctica. Evolution in these varied environments has produced an immense diversity of forms, including species that are microscopic and species whose members can grow to 18 m long (1.5 times the length of a school bus). In this chapter, we'll take a tour of the invertebrates, using the phylogenetic tree in Figure 33.1 as a guide. **Figure 33.2** surveys 23 phyla as representatives of invertebrate diversity.

▼ Figure 33.2 Exploring Invertebrate Diversity

Kingdom Animalia encompasses 1.3 million known species, and estimates of the total number of species range as high as 10–20 million. Of the 23 phyla surveyed here, 13 are discussed more fully in this chapter, Chapter 32, or Chapter 34; cross-references are given at the end of their descriptions.

Porifera (5,500 species)

A sponge

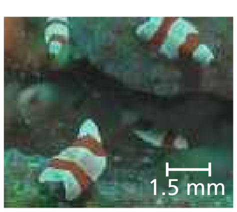
Animals in this phylum are informally called sponges. Sponges are sessile animals that lack tissues. They live as filter feeders, trapping particles that pass through the internal channels of their body (see Concept 33.1).

Cnidaria (10,000 species)

Cnidarians include corals, jellies, and hydras. These animals have a diploblastic, radially symmetrical body plan that includes a gastrovascular cavity with a single opening that serves as both mouth and anus (see Concept 33.2).

A jelly

Acoela (400 species)

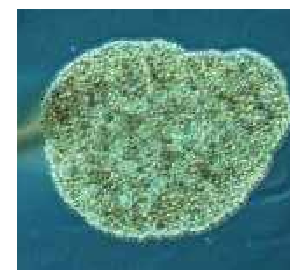


Acoela

Flatworms in this phylum have a simple nervous system and a saclike gut, and thus were once placed in phylum Platyhelminthes. Recent molecular analyses, however, indicate that Acoela is a separate lineage that diverged before the three main bilaterian clades (see Concept 32.4).

Placozoa (1 species)

The single known species in this phylum, *Trichoplax adhaerens*, doesn't look like an animal. It consists of a simple bilayer of a few thousand cells and reproduces by dividing into two individuals or by budding off many multicellular individuals. Placozoans are thought to be basal animals, but it is not yet known how they are related to Porifera, Cnidaria, and other early-diverging phyla.



A placozoan (LM) 0.5 mm

Ctenophora (100 species)

A ctenophore, or comb jelly

Ctenophores (comb jellies) are diploblastic and radially symmetrical like cnidarians, suggesting that both phyla diverged early from other animals. Comb jellies make up much of the ocean's plankton. They have eight "combs" of cilia that propel the animals through the water. When a small animal contacts the tentacles of some comb jellies, specialized cells burst open, covering the prey with sticky threads.

Lophotrochozoa

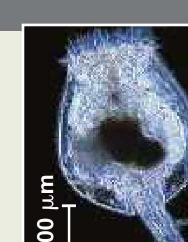
Platyhelminthes (20,000 species)

A marine flatworm

Flatworms in this phylum (including tapeworms, planarians, and flukes) have bilateral symmetry and a central nervous system that processes information from sensory structures. They have no body cavity or specialized organs for circulation (see Concept 33.3).

Syndermata (2,900 species)

This recently established phylum includes two groups formerly classified as separate phyla: the rotifers, microscopic animals with complex organ systems, and the acanthocephalans, highly modified parasites of vertebrates (see Concept 33.3).



A rotifer (LM)

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